

## CHAPTER 21

### SITE-SPECIFIC GROUND MOTION PROCEDURES FOR SEISMIC DESIGN

#### 21.1 SITE RESPONSE ANALYSIS

The requirements of Section 21.1 shall be satisfied where site response analysis is performed or required by Section 11.4.7. The analysis shall be documented in a report.

**21.1.1 Base Ground Motions.** An  $MCE_R$  response spectrum shall be developed for bedrock, using the procedure of Sections 11.4.6 or 21.2. Unless a site-specific ground motion hazard analysis described in Section 21.2 is carried out, the  $MCE_R$  rock response spectrum shall be developed using the procedure of Section 11.4.6, assuming Site Class B. If bedrock consists of Site Class A, the spectrum shall be adjusted using the site coefficients in Section 11.4.3 unless other site coefficients can be justified. At least five recorded or simulated horizontal ground motion acceleration time histories shall be selected from events that have magnitudes and fault distances that are consistent with those that control the  $MCE_R$  ground motion. Each selected time history shall be scaled so that its response spectrum is, on average, approximately at the level of the  $MCE_R$  rock response spectrum over the period range of significance to structural response.

**21.1.2 Site Condition Modeling.** A site response model based on low strain shear wave velocities, nonlinear or equivalent linear shear stress–strain relationships, and unit weights shall be developed. Low strain shear wave velocities shall be determined from field measurements at the site or from measurements from similar soils in the site vicinity. Nonlinear or equivalent linear shear stress–strain relationships and unit weights shall be selected on the basis of laboratory tests or published relationships for similar soils. The uncertainties in soil properties shall be estimated. Where very deep soil profiles make the development of a soil model to bedrock impractical, the model is permitted to be terminated where the soil stiffness is at least as great as the values used to define Site Class D in Chapter 20. In such cases, the  $MCE_R$  response spectrum and acceleration time histories of the base motion developed in Section 21.1.1 shall be adjusted upward using site coefficients in Section 11.4.3 consistent with the classification of the soils at the profile base.

**21.1.3 Site Response Analysis and Computed Results.** Base ground motion time histories shall be input to the soil profile as outcropping motions. Using appropriate computational techniques that treat nonlinear soil properties in a nonlinear or equivalent-linear manner, the response of the soil profile shall be determined and surface ground motion time histories shall be calculated. Ratios of 5% damped response spectra of surface ground motions to input base ground motions shall be calculated. The recommended surface  $MCE_R$  ground motion response spectrum shall not be lower than the  $MCE_R$  response

spectrum of the base motion multiplied by the average surface-to-base response spectral ratios (calculated period by period) obtained from the site response analyses. The recommended surface ground motions that result from the analysis shall reflect consideration of sensitivity of response to uncertainty in soil properties, depth of soil model, and input motions.

#### 21.2 RISK-TARGETED MAXIMUM CONSIDERED EARTHQUAKE ( $MCE_R$ ) GROUND MOTION HAZARD ANALYSIS

The requirements of Section 21.2 shall be satisfied where a ground motion hazard analysis is performed or required by Section 11.4.7. The ground motion hazard analysis shall account for the regional tectonic setting, geology, and seismicity; the expected recurrence rates and maximum magnitudes of earthquakes on known faults and source zones; the characteristics of ground motion attenuation near source effects, if any, on ground motions; and the effects of subsurface site conditions on ground motions. The characteristics of subsurface site conditions shall be considered either using attenuation relations that represent regional and local geology or in accordance with Section 21.1. The analysis shall incorporate current seismic interpretations, including uncertainties for models and parameter values for seismic sources and ground motions. If the spectral response accelerations predicted by the attenuation relations do not represent the maximum response in the horizontal plane, then the response spectral accelerations computed from the hazard analysis shall be scaled by factors to increase the motions to the maximum response. If the attenuation relations predict the geometric mean or similar metric of the two horizontal components, then the scale factors shall be 1.1 for periods less than or equal to 0.2 s, 1.3 for a period of 1.0 s, and 1.5 for periods greater than or equal to 5.0 s, unless it can be shown that other scale factors more closely represent the maximum response, in the horizontal plane, to the geometric mean of the horizontal components. Scale factors between these periods shall be obtained by linear interpolation. The analysis shall be documented in a report.

**21.2.1 Probabilistic ( $MCE_R$ ) Ground Motions.** The probabilistic spectral response accelerations shall be taken as the spectral response accelerations in the direction of maximum horizontal response represented by a 5% damped acceleration response spectrum that is expected to achieve a 1% probability of collapse within a 50-year period. For the purpose of this standard, ordinates of the probabilistic ground motion response spectrum shall be determined by either Method 1 of Section 21.2.1.1 or Method 2 of Section 21.2.1.2.

**21.2.1.1 Method 1.** At each spectral response period for which the acceleration is computed, ordinates of the probabilistic ground motion response spectrum shall be determined as the product of the risk coefficient,  $C_R$ , and the spectral response acceleration from a 5% damped acceleration response spectrum that has a 2% probability of exceedance within a 50-year period. The value of the risk coefficient,  $C_R$ , shall be determined using values of  $C_{RS}$  and  $C_{R1}$  from Figs. 22-18 and 22-19, respectively. At spectral response periods less than or equal to 0.2 s,  $C_R$  shall be taken as equal to  $C_{RS}$ . At spectral response periods greater than or equal to 1.0 s,  $C_R$  shall be taken as equal to  $C_{R1}$ . At response spectral periods greater than 0.2 s and less than 1.0 s,  $C_R$  shall be based on linear interpolation of  $C_{RS}$  and  $C_{R1}$ .

**21.2.1.2 Method 2.** At each spectral response period for which the acceleration is computed, ordinates of the probabilistic ground motion response spectrum shall be determined from iterative integration of a site-specific hazard curve with a lognormal probability density function representing the collapse fragility (i.e., probability of collapse as a function of spectral response acceleration). The ordinate of the probabilistic ground motion response spectrum at each period shall achieve a 1% probability of collapse within a 50-year period for a collapse fragility that has (1) a 10% probability of collapse at said ordinate of the probabilistic ground motion response spectrum and (2) a logarithmic standard deviation value of 0.6.

**21.2.2 Deterministic ( $MCE_R$ ) Ground Motions.** The deterministic spectral response acceleration at each period shall be calculated as an 84th-percentile 5% damped spectral response acceleration in the direction of maximum horizontal response computed at that period. The largest such acceleration calculated for the characteristic earthquakes on all known active faults within the region shall be used. The ordinates of the deterministic ground motion response spectrum shall not be taken as lower than the corresponding ordinates of the response spectrum determined in accordance with Fig. 21.2-1. For the purposes of calculating the ordinates

- (i) for Site Classes A, B or C:  $F_a$  and  $F_v$  shall be determined using Tables 11.4-1 and 11.4-2, with the value of  $S_1$  taken as 1.5 and the value of  $S_1$  taken as 0.6;
- (ii) for Site Class D:  $F_a$  shall be taken as 1.0, and  $F_v$  shall be taken as 2.5; and
- (iii) for Site Classes E and F:  $F_a$  shall be taken as 1.0, and  $F_v$  shall be taken as 4.0.

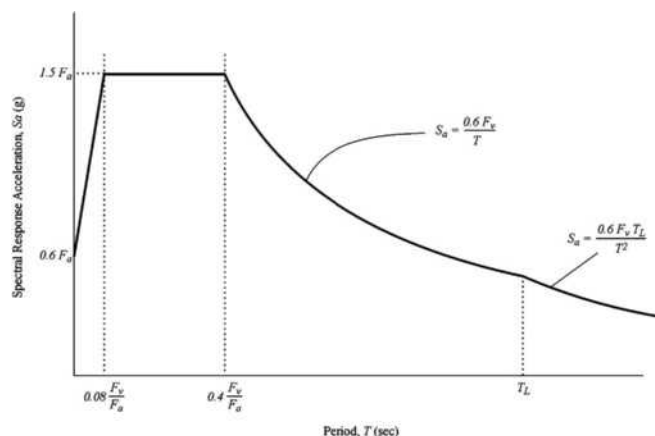


FIGURE 21.2-1 Deterministic Lower Limit on  $MCE_R$  Response Spectrum

**21.2.3 Site-Specific  $MCE_R$ .** The site-specific  $MCE_R$  spectral response acceleration at any period,  $S_{aM}$ , shall be taken as the lesser of the spectral response accelerations from the probabilistic ground motions of Section 21.2.1 and the deterministic ground motions of Section 21.2.2.

## 21.3 DESIGN RESPONSE SPECTRUM

The design spectral response acceleration at any period shall be determined from Eq. (21.3-1):

$$S_d = \frac{2}{3} S_{aM} \quad (21.3-1)$$

where  $S_{aM}$  is the MCE spectral response acceleration obtained from Section 21.1 or 21.2.

The design spectral response acceleration at any period shall not be taken as less than 80% of  $S_a$  determined in accordance with Section 11.4.6, where  $F_a$  and  $F_v$  are determined as follows:

- (i) for Site Class A, B, and C:  $F_a$  and  $F_v$  are determined using Tables 11.4-1 and 11.4-2, respectively;
- (ii) for Site Class D:  $F_a$  is determined using Table 11.4-1, and  $F_v$  is taken as 2.4 for  $S_1 < 0.2$  or 2.5 for  $S_1 \geq 0.2$ ; and
- (iii) for Site Class E:  $F_a$  is determined using Table 11.4-1 for  $S_1 < 1.0$  or taken as 1.0 for  $S_1 \geq 1.0$ , and  $F_v$  is taken as 4.2 for  $S_1 \leq 0.1$  or 4.0 for  $S_1 > 0.1$ .

For sites classified as Site Class F requiring site-specific analysis in accordance with Section 11.4.7, the design spectral response acceleration at any period shall not be less than 80% of  $S_a$  determined for Site Class E in accordance with Section 11.4.5.

**EXCEPTION:** Where a different site class can be justified using the site-specific classification procedures in accordance with Section 20.3.3, a lower limit of 80% of  $S_a$  for the justified site class shall be permitted to be used.

## 21.4 DESIGN ACCELERATION PARAMETERS

Where the site-specific procedure is used to determine the design ground motion in accordance with Section 21.3, the parameter  $S_{DS}$  shall be taken as 90% of the maximum spectral acceleration,  $S_a$ , obtained from the site-specific spectrum, at any period within the range from 0.2 to 5 s, inclusive. The parameter  $S_{D1}$  shall be taken as the maximum value of the product,  $TS_a$ , for periods from 1 to 2 s for sites with  $v_{s,30} > 1,200$  ft/s ( $v_{s,30} > 365.76$  m/s) and for periods from 1 to 5 s for sites with  $v_{s,30} \leq 1,200$  ft/s ( $v_{s,30} \leq 365.76$  m/s). The parameters  $S_{MS}$  and  $S_{M1}$  shall be taken as 1.5 times  $S_{DS}$  and  $S_{D1}$ , respectively. The values so obtained shall not be less than 80% of the values determined in accordance with Section 11.4.3 for  $S_{MS}$  and  $S_{M1}$  and Section 11.4.5 for  $S_{DS}$  and  $S_{D1}$ .

For use with the equivalent lateral force procedure, the site-specific spectral acceleration,  $S_a$ , at  $T$  shall be permitted to replace  $S_{D1}/T$  in Eq. (12.8-3) and  $S_{D1}T_L/T^2$  in Eq. (12.8-4). The parameter  $S_{DS}$  calculated per this section shall be permitted to be used in Eqs. (12.8-2), (12.8-5), (15.4-1), and (15.4-3). The mapped value of  $S_1$  shall be used in Eqs. (12.8-6), (15.4-2), and (15.4-4).

## 21.5 MAXIMUM CONSIDERED EARTHQUAKE GEOMETRIC MEAN ( $MCE_G$ ) PEAK GROUND ACCELERATION

**21.5.1 Probabilistic  $MCE_G$  Peak Ground Acceleration.** The probabilistic geometric mean peak ground acceleration shall be

taken as the geometric mean peak ground acceleration with a 2% probability of exceedance within a 50-year period.

**21.5.2 Deterministic  $MCE_G$  Peak Ground Acceleration.** The deterministic geometric mean peak ground acceleration shall be calculated as the largest 84th-percentile geometric mean peak ground acceleration for characteristic earthquakes on all known active faults within the site region. The deterministic geometric mean peak ground acceleration shall not be taken as lower than  $0.5 F_{PGA}$ , where  $F_{PGA}$  is determined using Table 11.8-1 with the value of PGA taken as  $0.5g$ .

**21.5.3 Site-Specific  $MCE_G$  Peak Ground Acceleration.** The site-specific  $MCE_G$  peak ground acceleration,  $PGA_M$ , shall be

taken as the lesser of the probabilistic geometric mean peak ground acceleration of Section 21.5.1 and the deterministic geometric mean peak ground acceleration of Section 21.5.2. The site-specific  $MCE_G$  peak ground acceleration shall not be taken as less than 80% of  $PGA_M$  determined from Eq. (11.8-1).

## **21.6 CONSENSUS STANDARDS AND OTHER REFERENCED DOCUMENTS**

See Chapter 23 for the list of consensus standards and other documents that shall be considered part of this standard to the extent referenced in this chapter.

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